What can waves tell us?

antocciphysics | TEACHER KEY | CPA

Learning target: Distinguish a signal's travel from the motion of its source.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 10; 3 course days later.*
The larger sphere has 1/3 of the smaller sphere's area per mass.

02. Explain a worked example (worked)
Pulse advances; marked segment moves locally and returns in the ideal model.

03. Look and explain (model)
No; the disturbance propagates while material oscillates locally in this model.

04. Build the solution (practice)
2 s.

05. Use the idea (reason)
Light travel delay; measurement uncertainty and motion during signal travel also matter.

06. Choose a method (mixed) *Review: Unit 5, Day 1; 12 course days later.*
Temperature change describes a thermal-state change (°C or K); heat is transferred energy (J). They are not the same quantity.

07. Exit check (exit)
Distance needs signal speed/time; velocity from Doppler needs wavelength/reference geometry; other justified pair.

08. STRETCH: Apply it another way (stretch)
Mark a rope segment or floating marker; compare its local motion with the advancing disturbance, using appropriate apparatus.

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antocciphysics | TEACHER KEY | HONORS

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 10; 3 course days later.*
Area factor 9, mass factor 27; A/m becomes 1/3. Survival also depends on transfer rates, fragmentation and conditions, not only energy per mass.

02. Explain a worked example (worked)
Pulse advances; marked segment moves locally and returns in the ideal model.

03. Represent and interpret (model)
No; the disturbance propagates while material oscillates locally in this model.

04. Solve and justify (practice)
3.0×10^{6} m; signal covers out-and-back distance.

05. Analyze the reasoning (reason)
Light travel delay; measurement uncertainty and motion during signal travel also matter.

06. Choose a method (mixed) *Review: Unit 5, Day 1; 12 course days later.*
No. Internal energy depends on amount of material and microscopic/material/phase structure as well as temperature.

07. Exit check (exit)
Second range 1.6 times first; delay does not establish mass.

08. STRETCH: Apply it another way (stretch)
Central 333.3 m/s; lower 300/0.95≈315.8, upper 300/0.85≈352.9 m/s.